Applied Corporate Finance

Week #2 – Initial Public Offerings

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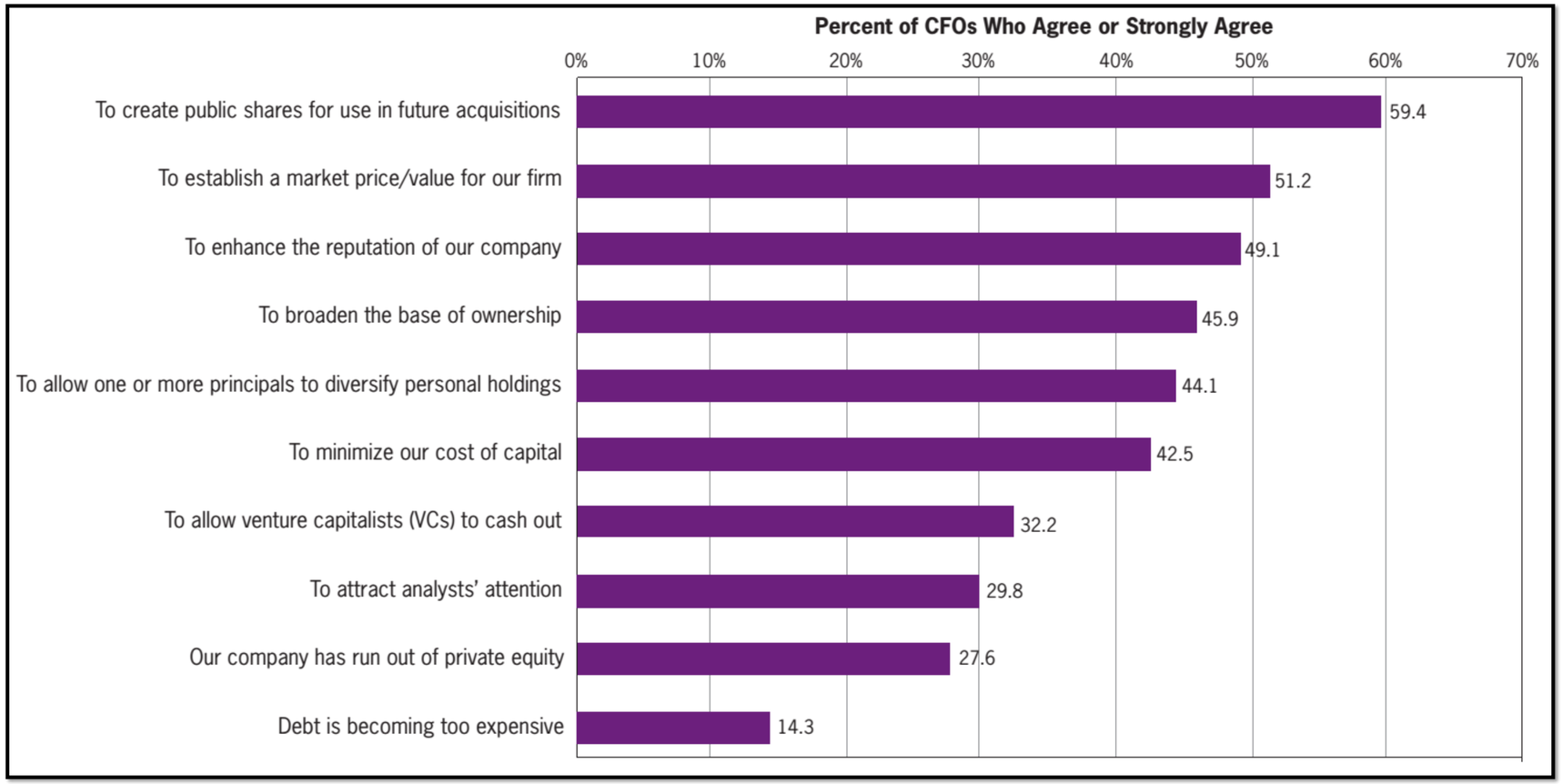
Plan of Attack

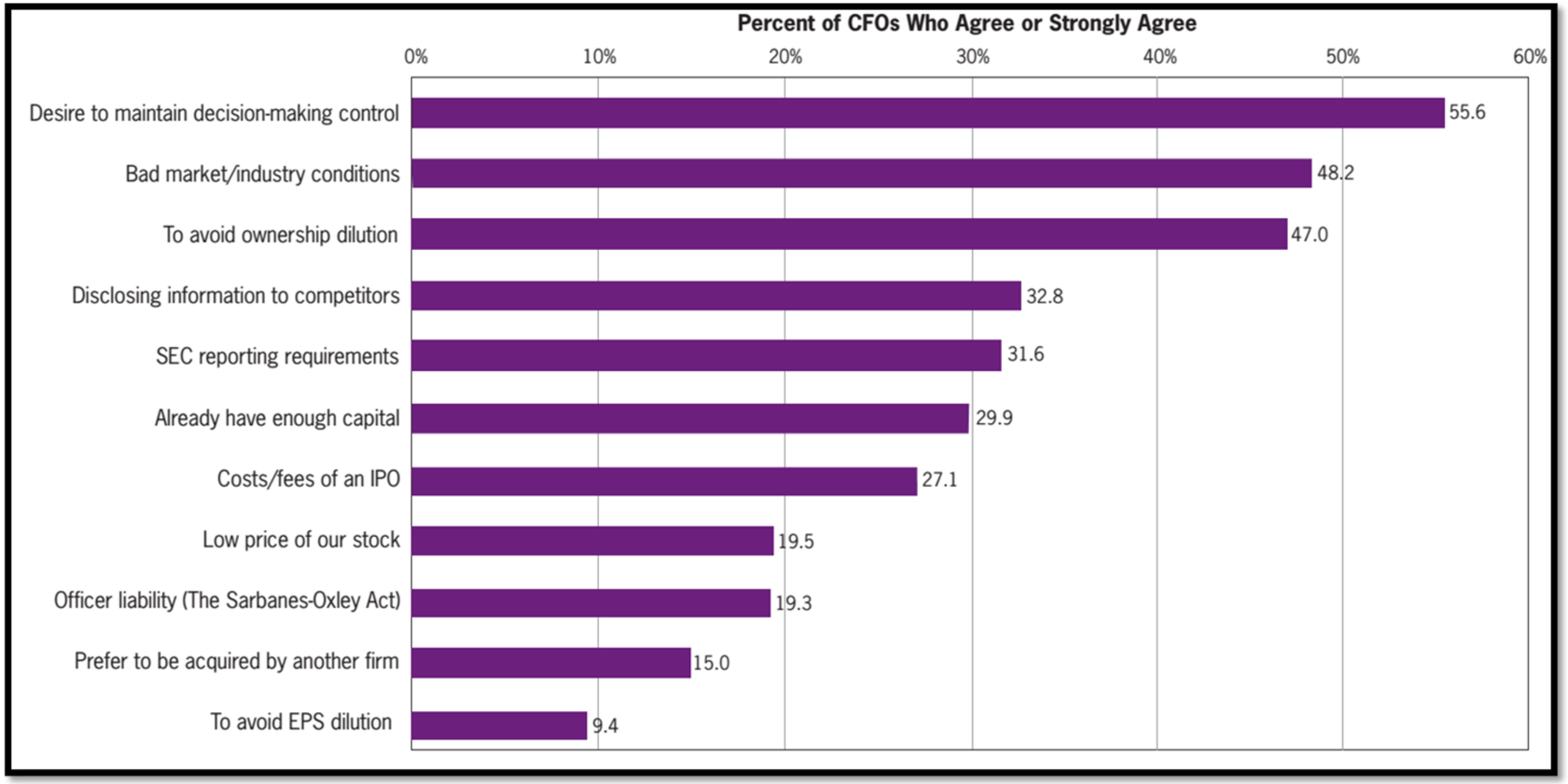
- **Why, When, and How to Go Public**
- **IPO Underpricing**
- **IPO Cyclicality**
- **Going private**
- **IPO Long Run Performance**
- **Appendix: The IPO Process**

Why Go Public?



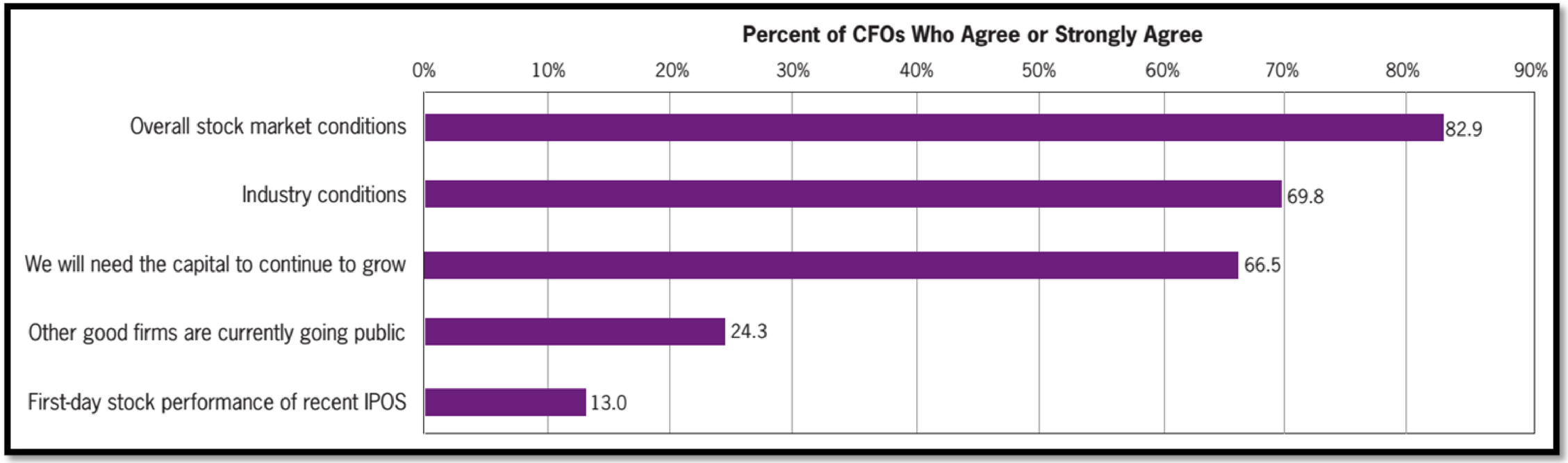
Source: Brau & Fawcett (2006)

Why Stay Private?



Source: Brau & Fawcett (2006)

When Do Firms Go Public?

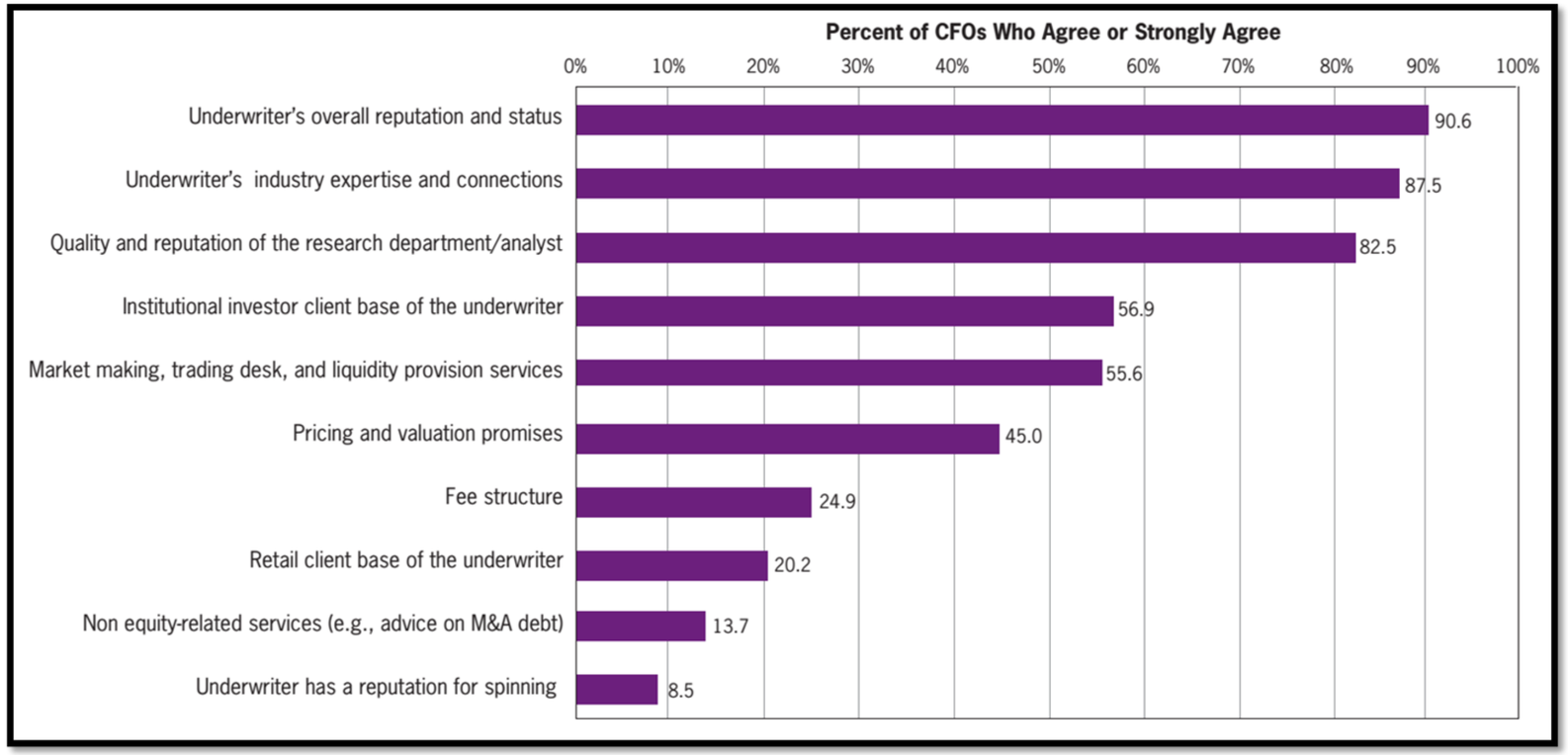


Source: Brau & Fawcett (2006)

The IPO Process

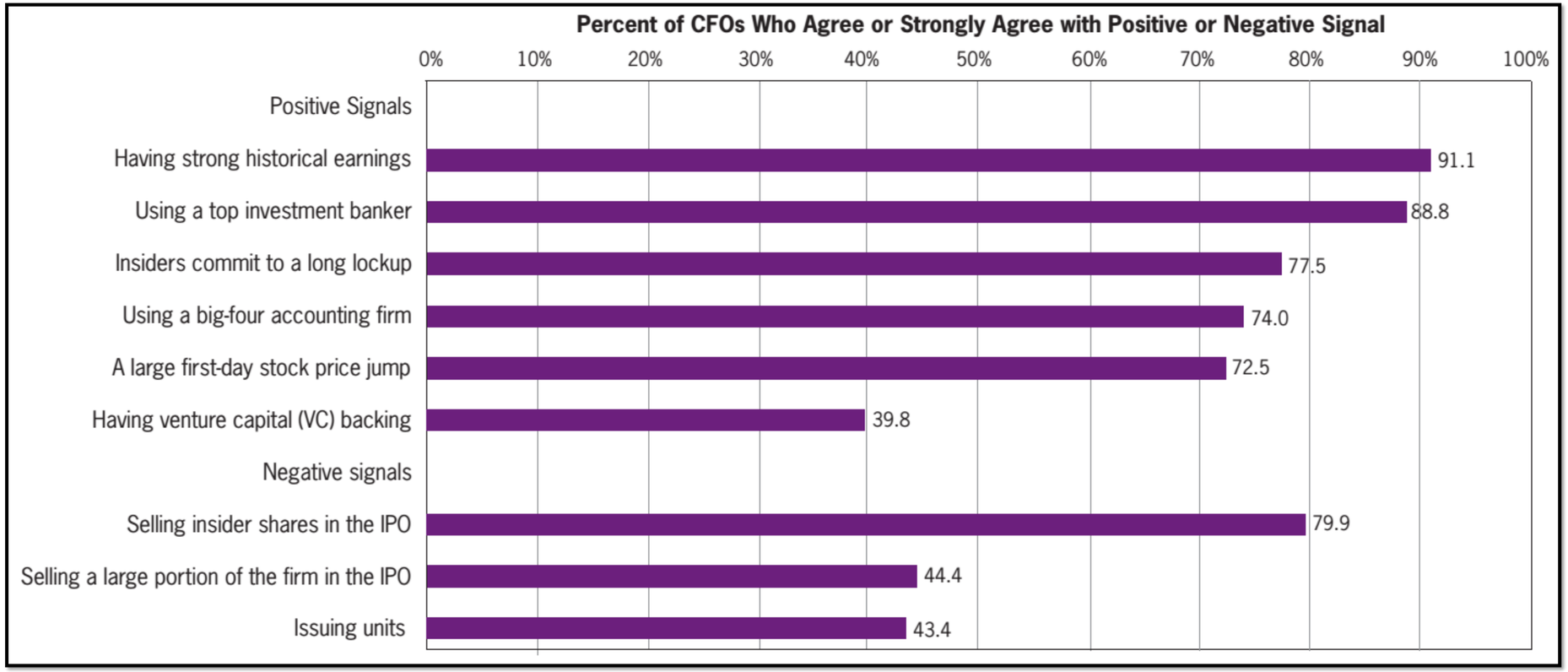
- Underwriter’s proposals
- Select Underwriter
- Registration Process (preliminary prospectus)
- Marketing and Road Show
- Pricing and Allocation

Criteria in Selecting an IPO Underwriter?



Source: Brau & Fawcett (2006)

Value of the IPO - Signals to Investors



Source: Brau & Fawcett (2006)

The Auction IPO Process

- Auction IPO basics
  - All bidders submit the number of shares they want to buy (quantity) and the price offer
  - The Underwriter aggregates all the information.
  - All shares are sold for the same price
  - That price is the highest bid that allows the underwriter to sell all the shares being offered.

- Examples:
  - Google's $1.67 Billion IPO in 2004 was done this way.
  - Morningstar also did an IPO auction.

- This method quite theoretically appealing, but still not common.

Special Purpose Acquisition Company (SPAC)

- A new (shell) company is formed with no assets
- The firm issues public shares with the goal of using the proceeds to acquire a target company
- The identity of the target company is typically not known or not disclosed at the time of the IPO
- Therefore, this is sometimes also referred to as “blank check company”
- After the issue, the proceeds are deposited in a trust fund until an acquisition is made
- If no acquisition within two years, money is returned to investors
- Investors can range from large institutions to retail investors
- This structure has gained tremendous popularity during COVID-19.

Pros and Cons of SPACs

Pros

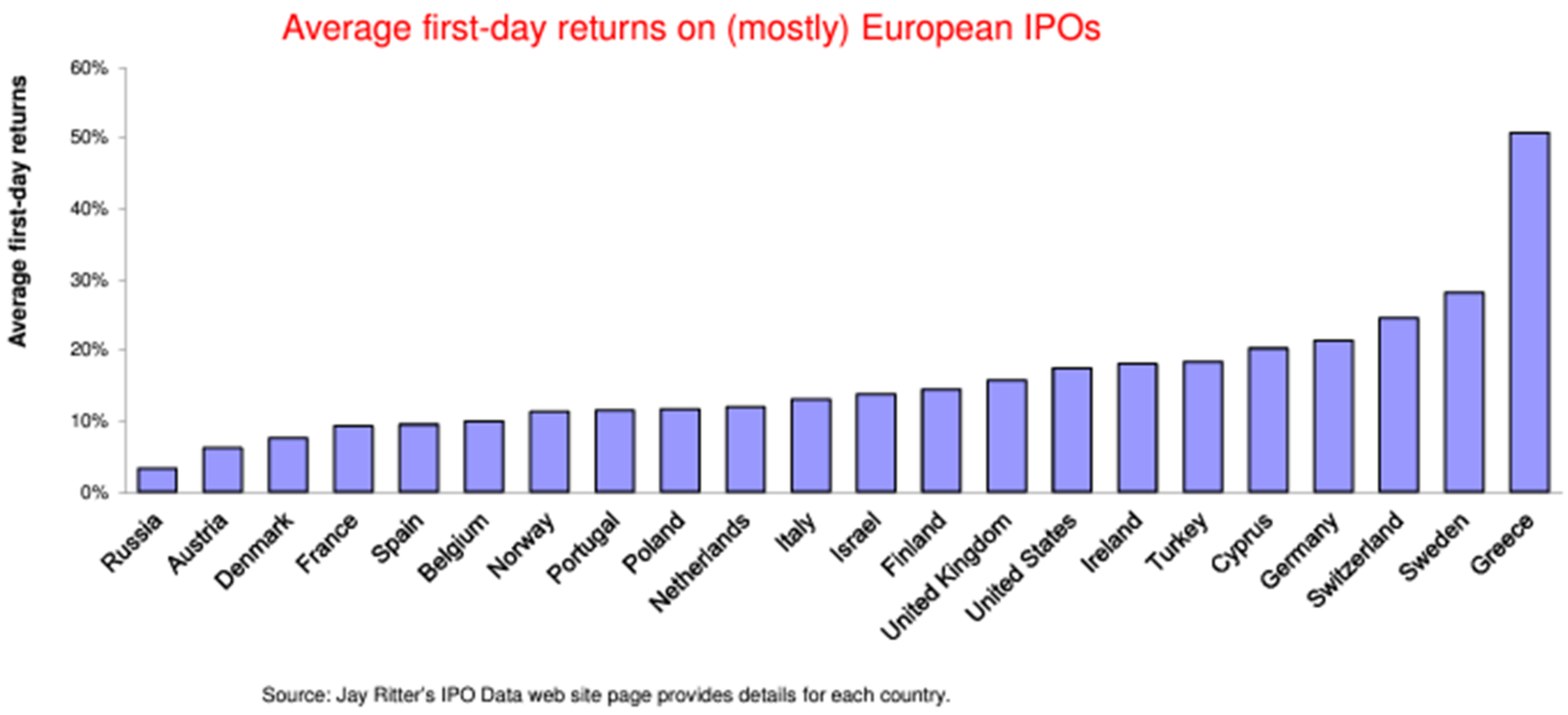
- Faster than a roadshow (less risk for the company with respect to changes in the market as the issue is being marketed)
- Only has to deal with an acquirer, not many potential investors
  - Maybe exponentiated due to COVID (?)
  - Maybe especially valuable for non-consensual companies
- Lower risk of public failure for the company
- Lower IPO regulatory and administrative burden (as the company that goes public is a simple “shell”)

Cons

- As an investor you don’t know what you are investing in
- In the past, these vehicles got a bad image due to sponsors profiting from investors money
- SPACs have historically less performance and a higher rate of bankruptcy

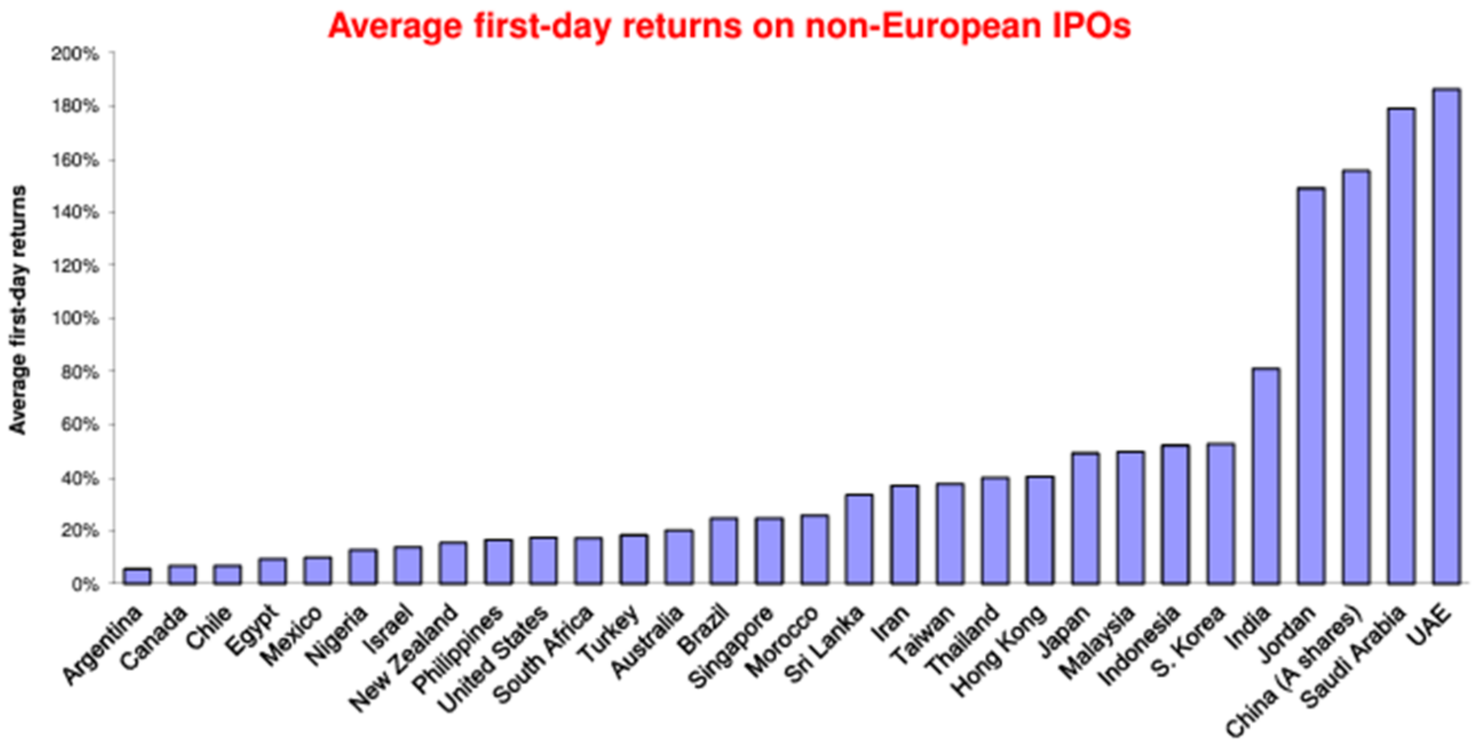
IPO Underpricing

IPO Underpricing Around the World



Source: Jay Ritter IPO Data, March 2024

IPO Underpricing Around the World



Source: Jay Ritter IPO Data, March 2024

Underpricing: Adverse Selection

- Informed and uninformed investors (Rock, 1986) – winners' curse model.
  - There are two classes of investors:
    - "informed investors" that know the true value of the IPO
    - "uninformed investors" that do not know the true value
  - There are not enough informed investors to buy all the shares in the market
    - both types need to participate.
  - Informed investors only buy when
    - Offering price ≤ true value
  - Uninformed investors buy always, but get more shares of bad IPOs because good ones are oversubscribed.

- For uninformed investors to break even there must be underpricing on the average IPO.

Underpricing: Information Acquisition

- **Information Transmission** (Benveniste & Spindt, 1989)
- Investors have information about the true value of the IPO firm
- The firm and the underwriter want to collect this information from the investors
- If you were the investor with superior information, would you share it with the underwriter? Why?

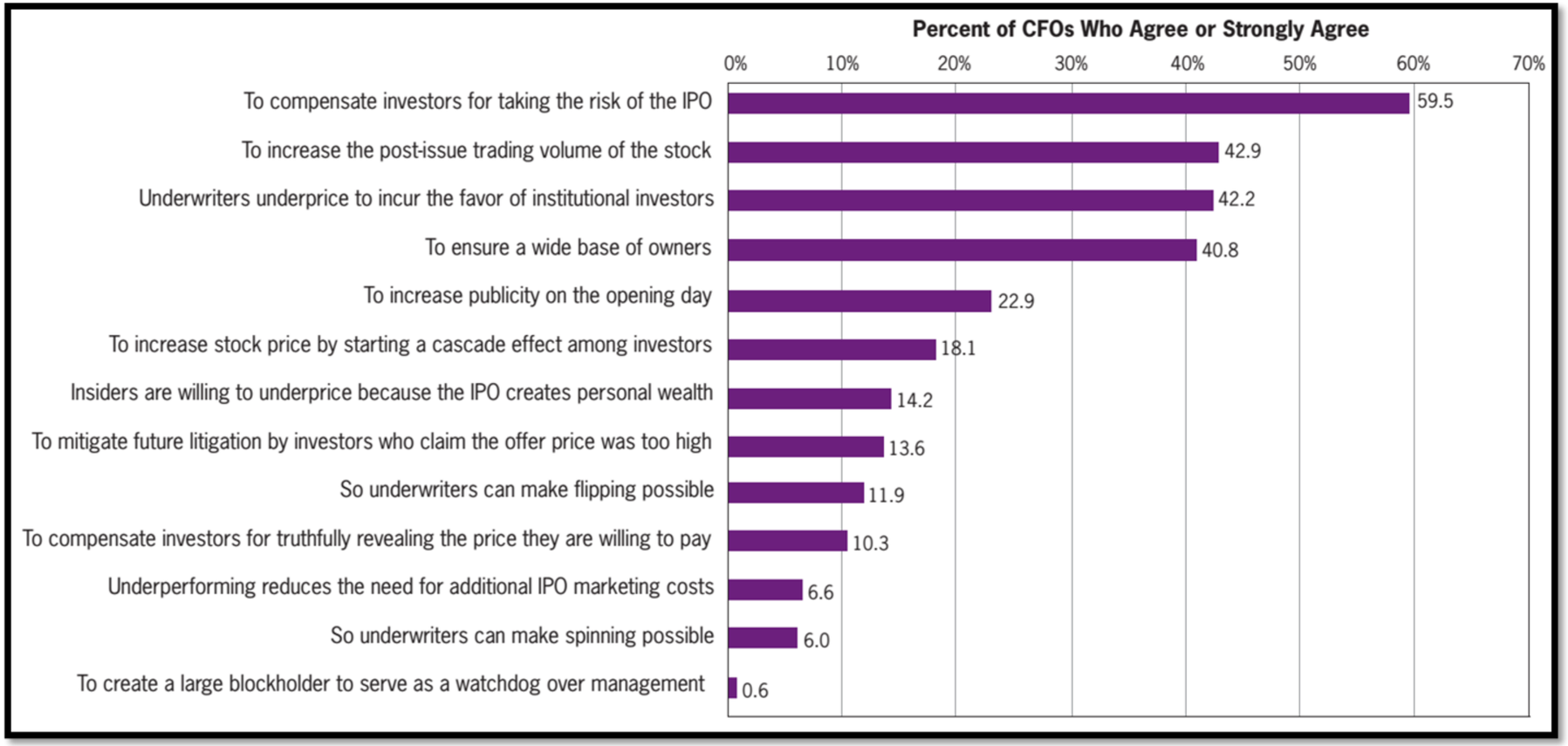
Underpricing: Information Acquisition

- In order for the **investors to share their information**, **underwriter must promise to reward them** for it.

- How can they do it?
  - **One way is to keep shares (relatively) cheap and allow those that provided the information to buy shares at the cheap price.**

- Takeaways from the Information Acquisition model:
  - Information acquisition during IPO process is key
  - Book building may be an efficient way to extract information from investors

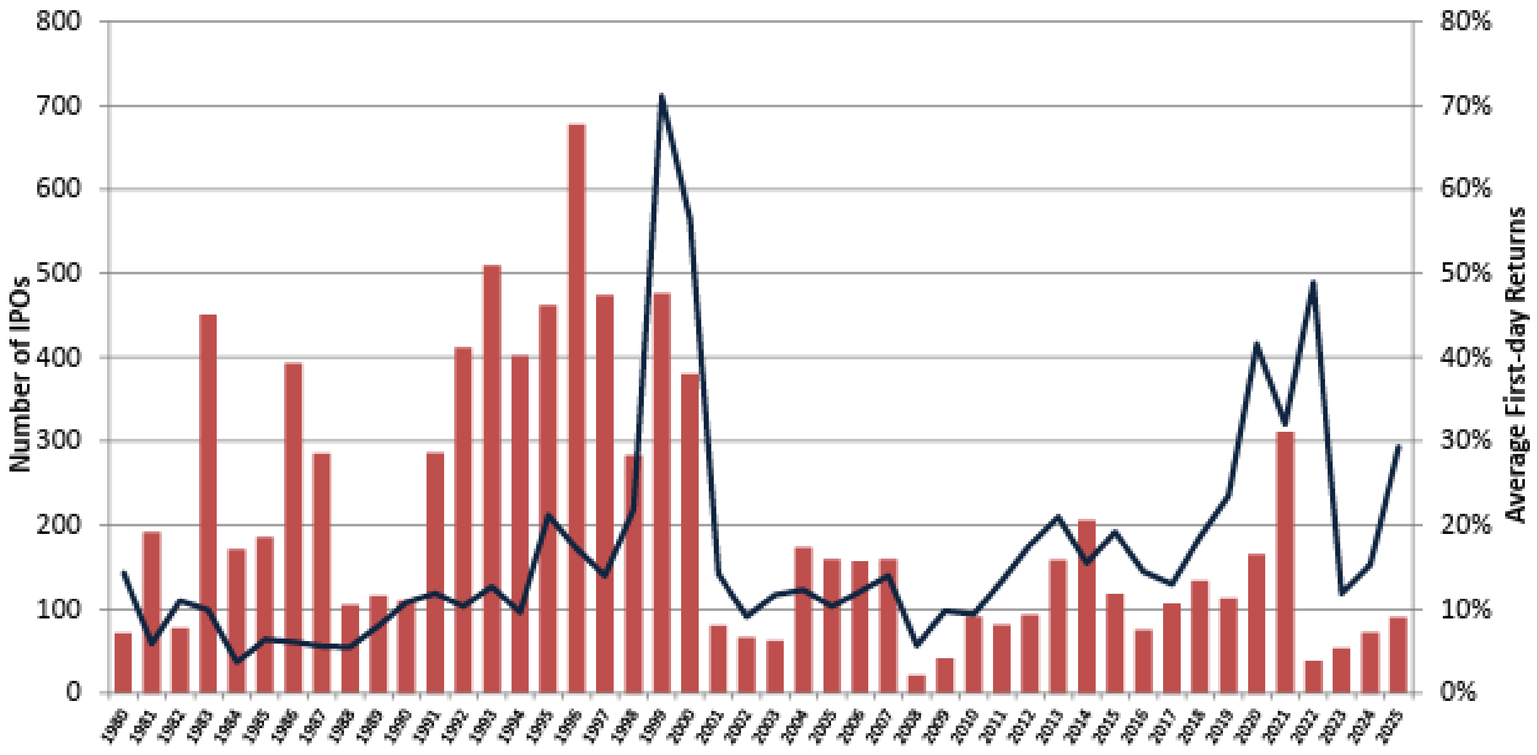
Underpricing: What Do CFOs Say?



Source: Brau & Fawcett (2006)

IPOs: Some Empirical Facts

Cyclicality



Number of IPOs

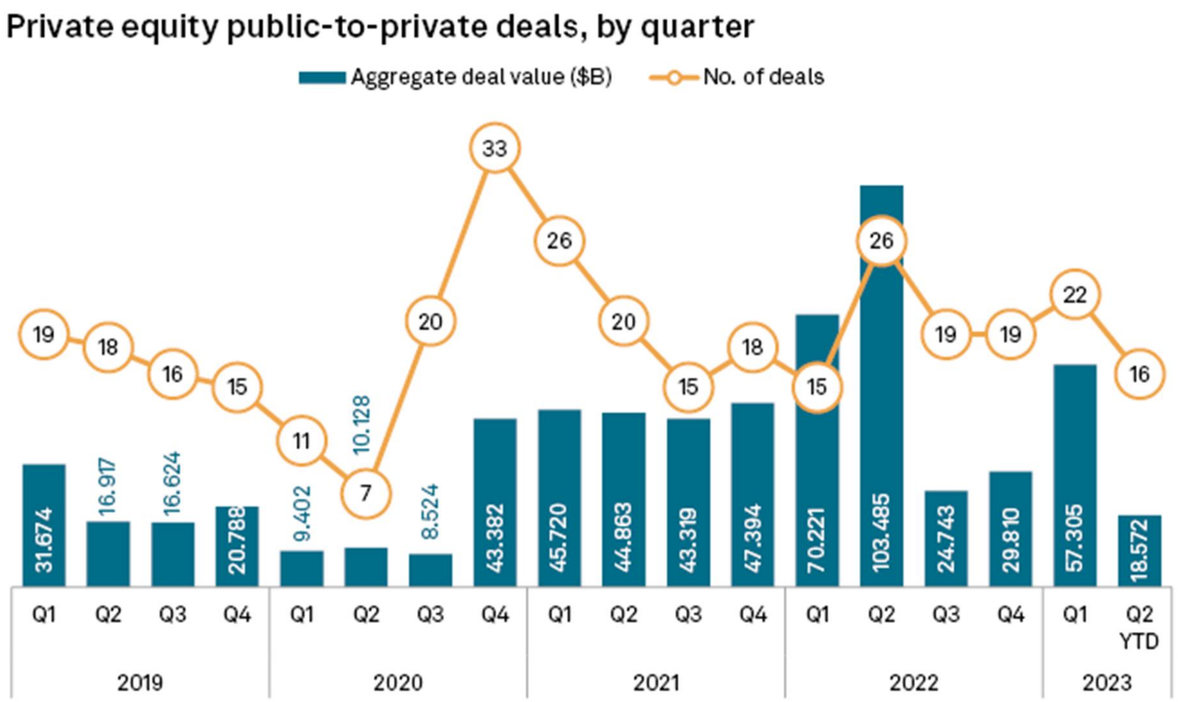
Average first day return

Source: Jay R. Ritter Initial Public Offerings in U.S.: Updated Statistics (January 2026)

Cyclicality

- Is the existence of cyclicality surprising?
  - When investment opportunities are good there should be more demand for funds.
- What is surprising is the magnitude of the swings.
- It is hard to imagine that the difference in growth opportunities would lead to a decline of 48% in the dollar volume of IPOs from 2000 to 2001.

Going private – the other way around



Data as of June 21, 2023.
Source: Preqin Pro.
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- To avoid costs.
- To facilitate restructuring actions.
- To incorporate in some other business (market power, synergies).
- To break-up business and sell in parts.
- …

Costs of Executing an IPO

- Typical underwriter gross spread is 7% of the issue price in the US, and around 3.75% in Europe
- The spread is not very sensitive to issue size. Why?
  - This is still a puzzle in the literature.
- Collusion?
  - Perhaps, but it does not seem to change much when market structure changes.
- Signalling?
  - Charging a lower fee could lead firms to believe the underwriter is of lower quality.
- This is a high cost relative to SEOs (Seasoned Equity Offerings) or bonds.

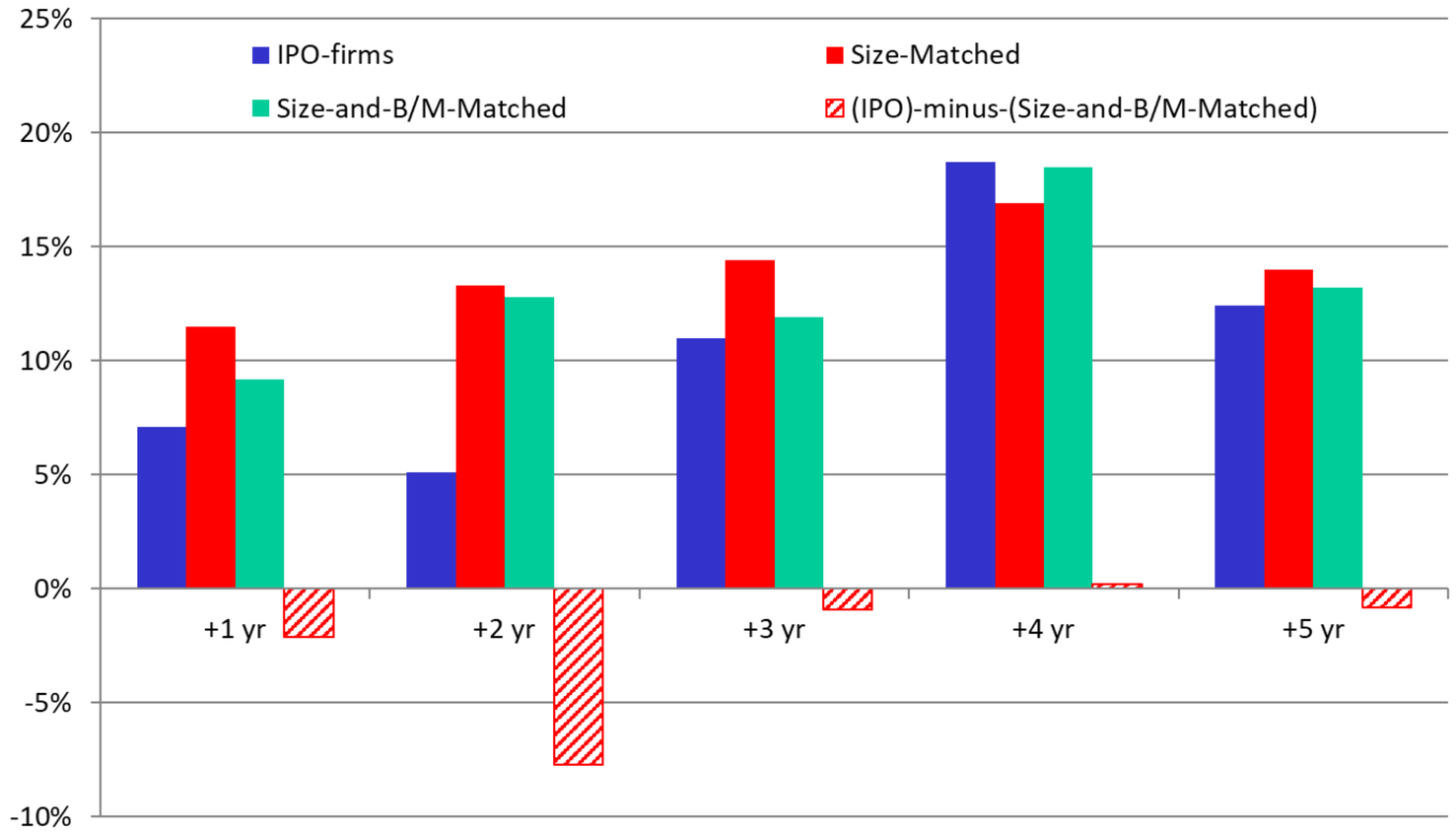
Long Run (under-) Performance

- A well-documented empirical fact is that IPO firms underperform in the long run.

- Ritter (2014) finds that from 1970-2012, IPO firms experience a 5-year annualized return of 10.7%
  - (not including the first-day return)

- 3.6 percentage points lower than similarly-sized matched firms.
- 2.2 percentage points lower than similarly-sized, similar style matched firms.

Long-Run (under-) Performance: 1980-2021



Source: Ritter (2023) Returns on IPOs during five years after issuing, for IPOs from 1980-2021

Pros and Cons of Going Public

- **Advantages:**
  - Greater liquidity for owners - typically done a while after IPO; liquidity can increase firm valuation
  - Better access to capital - e.g., follow-on equity offerings
  - Helps with interactions with customers and suppliers -- credibility
  - Creates currency for acquisitions and paying employees
  - Establishes observable price signal for contracting (e.g., CEO pay)

- **Disadvantages:**
  - Legal, accounting, investment bank fees are often high – up to 10% of amount raised
  - Dispersed shareholders - lower incentive/ability to monitor
  - Loss of control/potential for takeover
  - Information asymmetry could mean undervaluation
  - Legal requirements regarding information disclosure
    - Costly, both in terms of dollars and potentially competitive position.

Closing Comments about Going Public

- IPOs are a relatively expensive way to raise money
- Most IPOs are done through an underwriter that uses bookbuilding to establish the offering price for the firm.
- Underpricing -- average first-day returns of (Ritter Updated Statistics 2025):
  - 19% between 2001 and 2022
- There is significant cyclicality in the IPO market (still a puzzle)
- There are also companies going from public to private.
- Even when matched to similar firms, IPO firms seem to underperform in the long run.

Appendix: The IPO Process

The IPO Process

1. Underwriter's proposals

- Underwriters present proposals with:
  - Valuations
  - The extension of their services
  - The best type of security to issue
  - The offering price (usually a range)
  - Amount of shares (usually a range)
  - Estimated time frame for the market offering.

The IPO Process (cont.)

2. Select Underwriter

- This is a function of the reputation and expertise of the investment bank, particularly in firm's industry.
- A letter of intent is written
  - protects the underwriter against uncovered expenses in the event that the offer is withdrawn
  - Specifies the gross spread (typically 7%) and overallotment option (greenshoe option).
  - Remains in place until the Underwriting Agreement is executed at pricing
- Investment bank
  - Performs due diligence
  - Offering size analysis
  - Prepares marketing material and road shows
  - Arranges the syndicate

The IPO Process (cont.)

3. Registration Process

- Company files a registration statement to the regulating authority which is transformed into a preliminary prospectus.
  - E.g., Securities and Exchange Commission, state bodies.
- Meant to make sure that the public has all relevant information for their protection
- Forms the basis of the marketing material
- Also called the “Red Herring.”

The IPO Process (cont.)

4. Marketing and Road Show

- Firm's senior management and underwriters travel to visit potential investors to promote and explain their firm (2-3 weeks).
  - Mostly to institutional investors (e.g., mutual funds, pensions)
  - see http://www.retailroadshow.com/

- Underwriter engages in "bookbuilding."
  - Get preliminary indications from investors about their demand for shares at different prices
  - These "orders" are only indications – not legally binding (non-binding offers).
  - The road show is as much about gathering information from investors (their demand) as about disseminating information about the firm.

The IPO Process (cont.)

5. Pricing and Allocation

- Bank combines all the demand information from bookbuilding process.
- Firm managers and bankers discuss the final offer price and size.
  - The deal is usually “underpriced” (more on this later)
    - About 75% of firms see price increase on offering date.
- “Firm Commitment” offerings (as opposed to “Best Efforts”)
  - Bank commits to sell the shares at a set price (the night before the offering)
- Allocation decisions are made and the deal is executed.

The IPO Process (cont.)

6. Post-Offering Activities

- Bank implicitly assumes responsibility for providing price support, buying shares if order imbalances arise
  - Support period usually lasts about 20 days.
- Banks usually exercise over-allotment “greenshoe” option.
  - Allows the underwriter to issue more stock
  - Usually 15% of the original offer size
- Banks can manage risk by short-selling the greenshoe allotment size.
  - If issue is a success (and prices rise)
    - Underwriter exercises greenshoe option, covering their short
  - If the issue is not a success (and price falls)
    - Underwriters covers the short position by repurchasing the greenshoe allotment size in the open market (at a lower price that the short), thus supporting the price.

The IPO Process (cont.)

7. Post-Offering Relationship

- The firm-bank relationship often continues after the issuance.
- Future interactions include:
  - Analyst coverage
    - After “quite period” of 25 days ends
  - Market making
  - Financial advising